

MATH 31103

Intro to Abstract Algebra, I

Fall 2025

Professor John Bergdall
bergdall@uark.edu

MWF 9:40 - 10:30 AM
Classroom: WCOB 339

Overview

This course introduces abstract algebra. In algebra, we focus on mathematical structure. When are structures the same? Different? And how? We also focus on mathematical symmetry. How are objects internally built or organized?

Learning goals

Recognize algebra in diverse math contexts.

Study axioms of algebraic structures.

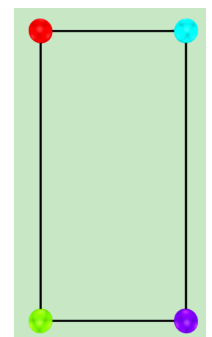
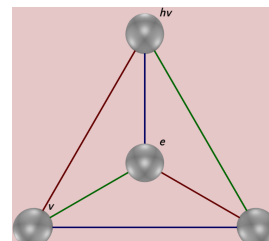
Classify foundational structures and examples.

Understand link between symmetry and algebra.

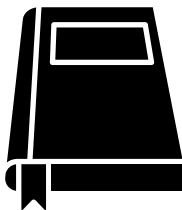
Try it out!

Visualizations of the *Klein 4-group* (first two) and the rectangle it symmetrizes. Explore more using <https://nathancarter.github.io/group-explorer/index.html>.

e	h	v	hv
h	e	hv	v
v	hv	e	h
hv	v	h	e



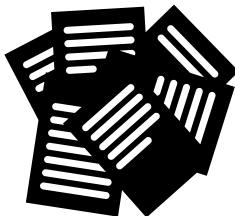
Basic resources



Textbook

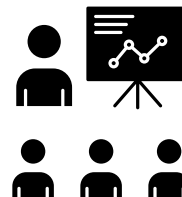
Abstract Algebra (4th ed.)
by Beachy and Blair
(Rent prints/digital: \$46/49*)

*If costs limit your course access, email me.



Course materials

Materials posted to Blackboard.
Assignments on Gradescope.
(Free)



Office hours (SCEN 353*)

Mon 2:30 - 3:30
Tue 1:30 - 2:30
Wed 12:00 - 1:00

*I have office chocolate for students.

Tasks and grades

Course tasks are divided into *Supporting Tasks* and *Performance Tasks*. Supporting Tasks represent class participation and readings. Performance Tasks represent chances to show your understanding of concepts and techniques. **Further details on pages 3-4.**



Supporting Tasks

Class Participation and Reading
Checks



Performance Tasks

Exercise Sets and Mastery
Assessments

Common turn-in

You will have two frequent kinds of assignments to turn-in. **Reading Checks** are due five minutes before the *start of class* (9:35 AM). **Exercise Sets** are due weekly on Wednesday at 11:59 PM.

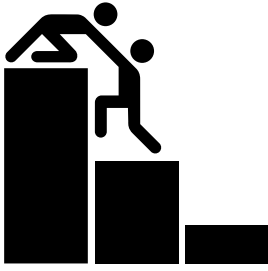
Revisions - Revisions reinforce learning. Except for a final exam, each task can be revised *once*.

- Revisions for Reading Checks are due weekly on **Wednesday at 11:59 PM**.
- Revisions for Exercise Sets are due roughly **36 hours** after the initial grades are returned.
- Exam revisions will take place in the week after initial grades are returned. (Details given in-semester.)

Late Tasks - Due dates help keep your learning on track. But things come up. *You may turn work in late as part of the revision process.* However, you lose your chance to revise the work.

Tip: start work early, so you always have something to turn-in. Imperfect work can be revised and will benefit from feedback!

Supporting Tasks



Class Participation gives you a voice in your learning. **Reading Checks** measure understanding of readings that frame class discussions.

Class Participation (variable points, 10 points total)

- **Attendance.** You are expected to regularly attend lecture.
- **Class surveys.** Three (3) surveys throughout the semester.
- **Class welcomes.** Two (2) meeting summaries read aloud to start class.

Details. One way students support learning is structural participation in the course.

- **Attendance.** Attending class is a key step in learning processes. You will earn **one point for 50%+** attendance and you will earn **two points for 75%+** for attendance.*
- **Class surveys.** At the start, middle, and end of the semester, I survey the course to reinforce what is working and address what is not. The first survey is not anonymous, but the others are. **Each survey is worth two points.**
- **Class welcomes.** A class welcome is a two paragraph summary of the previous lesson. You will write your summary with a partner, post it to Blackboard, and read it aloud start the next meeting. Sign-ups will be made in the first weeks of class. **Each welcome is worth one point.**

Reading Checks (equal weight per Check)

- **Two questions** per reading.
- **Initial responses** in timed and auto-graded assignment on Gradescope.
- **Revisions** will include complete reasoning, in writing.



Details. Reading Check questions are *multiple choice* or *select all*.

- Before the Check, you will know a prompt for the question but not its precise version.
- You take the Check prior to a meeting (due 9:35 AM). I use the questions frame the class discussion.
- Checks are open book/notes. You *may not* use generative AI. **See page 15 for AI guidelines.**
- Revision is possible if you do not get a question initially correct!

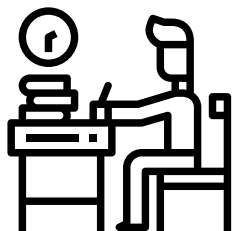
Revision specifications. Here is how you will “revise” your Reading Quizzes. **See page 12-13 for examples.**

- **Multiple choice.** Identify the correct answer along with a detailed explanation for why it is correct.
- **Select all.** First, identify the correct and incorrect options. Second, explain in detail why the correct ones are correct *and* why the incorrect ones are incorrect. You must do both steps!
- Complete and accurate revisions **earn back full points.** Some explanations may earn a half point.

* My expectation for “full” attendance is usually you miss a day every other week ($\approx$ 80% attendance). I am setting expectations a little lower because I myself have four days of planned research travel. Fair is fair, amirite? There are 41 class meetings. You can miss 10 and still earn full attendance points. Please read the Expanded Policies for a reminder on University-recognized reasons for absence.

Performance Tasks

Exercise Sets are first shots at concepts and techniques. **Mastery Assessments** are timed in-class assessments that measure broad proficiency.



Exercise Sets (equal weight per Set)

- **A few questions** per section of Beachy-Blair, due Wednesday nights.
- **Turned in** electronically through Gradescope.
- **Collaborate!** As long as you attribute credit to your source(s).

Mastery Assessments

- **Timed assessments** focused on vital learning goals and skills. Consists of a mix of calculations and conceptual prompts.
- **In-class midterm:** Wednesday 10/08, on Secs 1.1-3.3.
- **Final exam:** Monday 12/08, 10:15 AM - 12:15 PM in WCOB 339.



Details. Exercise Sets and Mastery Assessments are carefully designed based on learning goals.

- Some problems on Exercise Sets will be modeled during lecture. Others may require further analysis. Regardless, I post solutions to Blackboard (after revisions are complete, see below).
- For Mastery Assessments, I post reviews and practice exams one week prior to the assessment. We will also hold review time during class meetings. (See the schedule on page 9.)
- On Exercise Sets, you *may not* use generative AI to answer questions. **See page 15 for AI guidelines.**
- Revision is possible if you do not get a question initially correct!

Revision specifications. First, when revising performance task questions, your new scores replace your old score! The revisions themselves are due at 11:59 PM on Mon-Thu no earlier than 36 hours after the initial grades are returned. (Do not stress about the complex rule. I'll announce clear deadlines as they come up.) **See page 14 for examples.**

- **Exercise Sets.** To revise a question on a Exercise Set, you should turn in a new solution along with an indication of your prior score.* We will assess the work as if we never saw the prior answer.
- **Midterm Mastery Assessment.** Revising your midterm Mastery Assessments is a two-step process.
 - **Step #1.** Turn in new and complete solutions to the problems you aim to correct. These explanations are due within the 36-hour revision window.
 - **Step #2.** If I judge your explanations *completely* accurate (i.e. worthy of 6/6), I give you a second assessment comprised of problems similar to the ones you aim to revise. You will take this assessment in the Math Resource and Teaching Center (MRTC).

* Your prior score is needed to simplify the logistics of revisions. Please, please, include your prior score so I do not have to look it up.

Performance Task Grading

Your Performance Tasks will be graded using a 6-point rubric. The rubric will be strictly applied. Remember, revisions are always possible!

Examples of this rubric is applied are on pages 10-11.

Details Excelling in my 3xxxx-level courses requires both understanding concept and strategy as much as mechanics and computations. It also requires presenting answers in easy to read fashion, with clear explanations. Our rubric emphasizes all of these points.

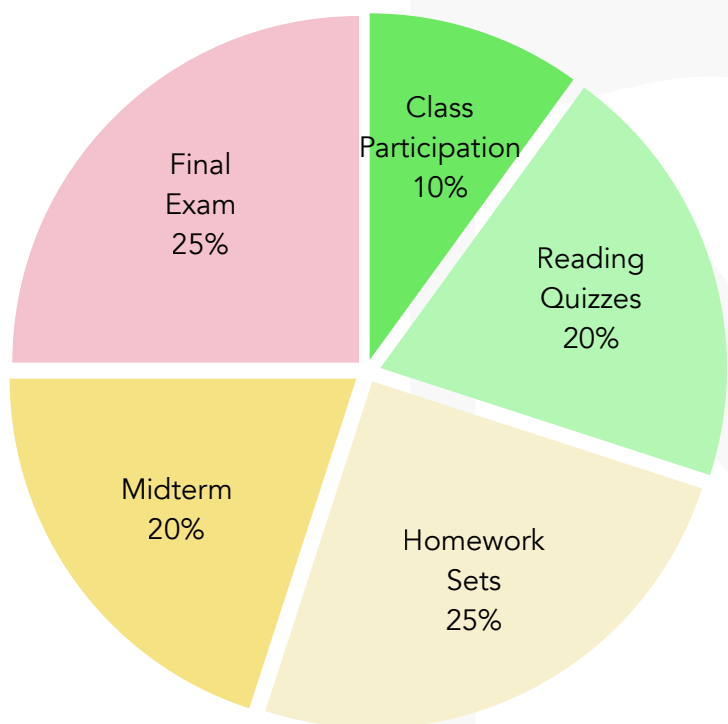
	0 points	1 point	2 points
Concepts or strategy (up to 2 points)	The answer misidentifies or misuses crucial concept(s) or applies unproductive strategies.	The answer identifies and uses a crucial concept or attempts a productive strategy.	The answer identifies or applies <i>all</i> crucial concepts and succeeds in carrying out a productive strategy.
Mechanics or computation (up to 2 points)	The answer has significant or recurring issues with its mechanics or computation.	The answer shows proficiency in some mechanics and computation.	The mechanics and computations in the answer are flawless (and of the intended difficulty level*).
Grade-ability (up to 1 point)	The solution is not for the right problem, its writing is illegible, or grading is impossible for another reason.	The solution is for the right problem and the writing is neat enough to assess the work.	* "...intended difficulty level" means you don't simplify the problem to do an accurate calculation. See Solution #4 in the example on page 11.
Presentation (up to 1 point)	The solution omits vital writing necessary to explain the logic and reasoning needed for mathematical discourse.	The solution is written in mathematical discourse, with complete sentences giving all necessary explanations.	

Final exam caveat. There is no revision possible on the final exam. So, "...all crucial..." and "flawless" in the 2-point criteria above will be suitably relaxed during the final exam.

Final grade

Your course grade will be derived from your Task scores according to the percentages in the pie chart below. The weighted scores determine your final grade according to the table on the right. *I reserve the right to make the gradelines more lenient.*

Task-by-Task weights for final grade



Gradelines

Grade	Minimum weighted score
A	90%
B	80%
C	70%
D	60%

Rubric interpretation. The Performance Tasks (70% of your grade) are graded on a 0-6 scale and yet the gradelines are on a 90/80/70/60 scale. Here is one way to think about the Performance Tasks scores.*

- **A-level Performance Tasks = average 5.4.** So, half the time your work is excellent (earning 6's) and the other half there is at least excellence in concepts or mechanics (earning 5's).
- **B-level Performance Tasks = average 4.8.** So, nearly all the time your work is at least excellent in one of mechanics or concepts but maybe not the other (earning 5's).
- **C-level Performance Tasks = average 4.2.** Your work routinely shows proficiency in mechanics and concepts (earning 4's).

* This assumes you do the right problems and present your explanations in a clear fashion (to earn the Grade-ability and Presentation points).

Tips for success

I designed the course for you to pass. The revision process puts you in control of your grade. Everyone will need some help at some point in the semester. So, here are a few tips and resources you can pursue.

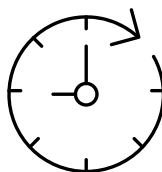


Message!

Write for help.



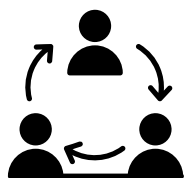
Ask! Me early and often.



Budget! Take your time.



Read! The text is a guide.



Collaborate!

With peers.

More tips (roughly in an order I follow)

- Take a break. Walk away.
- Explain to someone why you are stuck.
- Work out a simpler example.
- Search the internet.
- Ask an AI assistant. **See page 15 for AI guidelines.**
- Keep going!
- Tutoring is available in the Student Success Center in CORD.

Further Resources

Appointments

Office hours will not always fit your schedule. Email me to set appointments. In your initial message, please explain (i) what you want to talk about, (ii) *your* availability and (iii) if meeting in person is important (Zoom is possible).



Email

Your questions are important. Email me or post to Blackboard. (I may ask you to do this anyways.) Responses may take 24 hours, but send a reminder if I do not reply.

Access

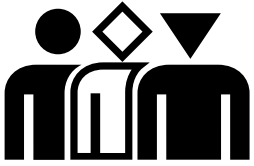
I want you to pass this class, with success. If you find resources not being provided, tell me and I will work with you to fix the issue. For some students, the Center for Educational Access (cea.uark.edu) coordinates student accommodation requests. Please request your accommodation letter early semester and meet with me to discuss it.



Class formation

Names and pronouns

You deserve to be called how you want. Let me know your preferred name and pronouns any time. When meeting one another, reintroduce yourselves if necessary. There is no reason to not know how someone is called.



Diversity

I pledge attention to your identity and experience, regardless of age, background, beliefs, ethnicity, gender identity and expression, national origin, racial identity, religious beliefs, sexual orientation, and any other visible or non-visible categories. You are expected to do the same for me and your peers.

Academic integrity

I trust my students to follow (i) the University of Arkansas Academic Integrity Policy and (ii) my own explicit assignment instructions. When you have questions about a boundary or rule, ask me rather than assuming you know the answer.



Technical policies

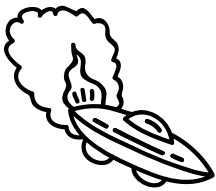
Inclement weather

If the University closes due to inclement weather, our course will meet online. The meeting will be recorded, as dictated by University policy. Any assessment or due date will be postponed until the next day the University is open.



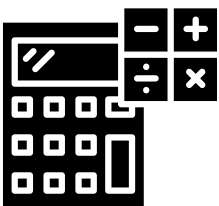
Class cancellation (emergencies)

If class is cancelled on short notice, I will work with the math office to try to find a substitute. If a substitute is found, class will be held. If not, alternative learning materials and assessments will be supplied. Regardless, an announcement will be sent promptly via email and posted to Blackboard.



Recordings and devices

I record class meetings using in-room recording systems and notes are live streamed via the OneNot (part of Office 365). If you use your devices (phone, tablet, etc.) to participate in class, you are expected to use those devices productively. Please tell me if someone's device is distracting you from your learning.



Calculators

A simple non-programmable calculator is not required. However, they are allowed and may give you confidence in your calculations during exams.

Schedule

	Wk.	Dates	Topic(s)	Reading
Recognize algebra in diverse math contexts.	1	Aug 18-22	Greatest common divisors, Smith normal form	Sec. 1.1*
	2	Aug 25-29	Prime factorization and modular arithmetic	Sec 1.2-1.4
	3	Sep 3-5	Abstract algebra of permutations	Sec 2.3
	4	Sep 8-12	Introduction to groups	Sec 3.1
Study axioms of algebraic structures.	5	Sep 15-19	Abstract and concrete examples (abelian and cyclic groups, unit groups, and direct products of groups)	Sec 1.4, 3.1-3.3
	6	Sep 22-26	Subgroups and examples	Sec 3.2, 2.3
	7	Sep 29-Oct 3	Lagrange's theorem: applications and proof	Sec 3.2
	8	Oct 6-8 Oct 10	Midterm review and exam Introduction to isomorphisms	Sec 3.4
Classify foundational structures and examples.	9	Oct 15-17	More isomorphisms and classification of cyclic groups	Sec 3.4, 3.5
	10	Oct 20-24	Group classification, presentations, homomorphisms	Sec 3.6-3.7*
	11	Oct 27-31	Kernels, normal subgroups, and intro to factor groups	Sec 3.7-3.8
	12	Nov 3-7	Factor groups, finite abelian groups, simple groups	Sec 3.8, 7.5, 7.7
Understand link between symmetry and algebra.	13	Nov 10-14	Group actions and symmetries of polyhedra	Sec 7.3*
	14	Nov 17-21	Applications of group actions	Sec 7.4*
	15	Nov 24	Applications..., cont.	Sec 7.4
	16	Dec 1-3	Review for final exam	—

*Additional handouts will be provided.

Icon Credits

All icons provided under Creative Commons (CC BY 3.0) licenses via www.nounproject.com. Specific creators listed below.



textbook by Athok



papers by Rean-me



presentation by iBrandify



support by lakonicon



creative by Chaya Kurniawan



reading-book by Anconer Design



collaborate by Vectors Point



homework by Chattapat



dissertation by WEBTECHOPS



victory by Adrien Coquet



texting by Luis Prado



confused-man by ages raharjo



clock by Dinosoftlabs



read by ahmad



teamwork by Gilbert Bages



appointment by Aneeqe Ahmed



emails by lakonicon



access by Nur Faizah



name-tag by icon_background



diversity by Luis Prado



academic by MRK



snowman by Smalllike



accident by Hey Rabbit



device by Putri Amaliya



calculator by kolifah

Performance Task Rubric

Here we provide examples for how Performance Tasks are graded. We chose problems from calculus, hoping that you will get the general idea without being confused about the mathematics.

Question.

Determine the area between the curve $y = \sin(x)$ and $y = \cos(x)$ over the x -range of $[0, \pi/4]$.

As a grader, I anticipate a student will convert the question into an integral calculation by subtracting one function from the other. Then, they will carry out the integral calculation by finding antiderivatives.

Solution #1.

$$\cos(x) \geq \sin(x) \text{ on } [0, \pi/4]$$

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = \sin(x) + \cos(x) \Big|_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414 \dots$$

Solution #2. We find the area by calculating an integral. Since $y = \cos(x)$ lies above $y = \sin(x)$ on $[0, \pi/4]$ (I graphed it), the relevant integral is:

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx.$$

Now we calculate:

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = \sin(x) - \cos(x) \Big|_0^{\pi/4} = 0 - 1 = -1.$$

Since areas are positive, the area between the curves is 1.

Grading solution #1: 4/6.

- *Concept.* 2/2. The solution finds an integral, after identifying which function to subtract from the other.
- *Mechanics.* 1/2. The integral is calculated completely accurately but the inequality is not explained.
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 0/1. There is no writing that indicates the goal, outcome, or approach of the solution. (This include the mysterious " $\cos(x) \geq \sin(x)$ " just floating on the page.)

Important point: the question asks for an *area* between two curves. The solution just calculates an integral. The missing explanation is how to connect the two.

Grading solution #2: 4/6.

- *Concept.* 1/2. The solution finds an integral to calculate. However, if we believe the setup is right then there is a conceptual error in the calculation turning up the answer -1. That is inconsistent conceptual knowledge.
- *Mechanics.* 1/2. The integral is not calculated accurately because there is a mistake in the antiderivative of $\sin(x)$. (Explaining you graphed the functions is fine, in this case, so long as you are right.)
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 1/1. The writing narrates in a clear way how the question is being answered.

This may take getting used to, but I consider conceptual gaps to also include situations where an answer is given that is inconsistent with some of the conceptual and foundational knowledge surrounding the question.

Performance Task Rubric

Here we provide examples for how Performance Tasks are graded. We chose problems from calculus, hoping that you will get the general idea without being confused about the mathematics.

Question.

Determine the area between the curve $y = \sin(x)$ and $y = \cos(x)$ over the x -range of $[0, \pi/4]$.

As a grader, I anticipate a student will convert the question into an integral calculation by subtracting one function from the other. Then, they will carry out the integral calculation by finding antiderivatives.

Solution #3. We find the area by calculating the integral

$$\int_0^{\pi/4} \sin(x) - \cos(x) dx = 1 - \sqrt{2} \approx -0.414\dots$$

Since areas are positive, the answer is 0.414...

Solution #4. The area is given by

$$\sin(\pi/4) - \cos(\pi/4) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0.$$

Grading solution #3: 5/6.

- *Concept.* 1/2. The solution finds an integral but it has misidentified the difference to integrate and there is no indication the solve realizes the issue.
- *Mechanics.* 2/2. The integral is calculated just fine.
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 1/1. The writing narrates in a clear way how the question is being answered and what the outcome is.

You may want to compare with Solution #2. The solutions are similar. Here, the mechanics are perfect and there they are not. In conceptual terms, both solutions have issues. In this #3, you might argue there is more conceptual gaps. That's okay. My rubric rewards flawless aspects of the work while being generous with mistakes. Similar mistakes do not count double in my grading scheme. Read carefully how you earn 1/2 on the concept or mechanics points.

Remember: your professors have enormous agency in how they design grading. Grading is not objective, even if someone has told you otherwise. To mitigate that challenge, my rubric is designed to prioritize being transparent and upfront.

Grading Solution #4: 3/6.

- *Concept.* 0/2. The solution does not point to a productive strategy to answer the question as given. (Plus, it is a conceptual issue to get area = 0.)
- *Mechanics.* 1/2. The calculation is literally correct but it overly simple.
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 1/1. The answer is clear in what it claims has happened.

Read. Check Revisions

Your Reading Checks will feature both *multiple* choice and *select all* questions. Depending on which type you revise, your revision will ask more or less of an explanation. Let's see how this look in practice.

Question. (Multiple choice) Which of the following indefinite integrals are calculated accurately?

(a) $\int \sin(x) dx = \cos(x) + C.$

(b) $\int \cos(x) dx = \sin(x) + C.$

(c) $\int \tan(x) dx = \sec^2(x) + C.$

Revision (1/1). The integral (b) is accurate because I looked it up on pg. 431. Since this is a multiple choice question, I do not need to address options (a) and (c).

Tip: You can justify your statement from a trusted source. (By the way, pg. 431 is just a random number.)

Question. (Select all that apply) Which of the following indefinite integrals are calculated accurately?

(a) $\int \cos(x) dx = \sin(x) + C.$

(b) $\int \sin(x) dx = \cos(x) + C.$

(c) $\int \tan(x) dx = \sec^2(x) + C.$

Revision (1/1). The integral (a) is accurate but (b) and (c) are not. The answer (a) is true because I looked it up on pg. 431 of our book. Next, (b) and (c) are false because it also says $\int \sin(x) dx = -\cos(x) + C$ and $\int \tan(x) dx = \ln(|\sec(x)|) + C.$

Tip. To show something is false, you can provide a corrected statement.

Question. (Select all that apply) Suppose $y = f(x)$ and $y = g(x)$ are two curves lying over x in $[a, b]$. Which of the following statements are accurate?

(a) The area between the two curves is always non-negative.

(b) The integral $\int_a^b f(x) - g(x) dx$ represents the area between the two curves.

(c) The integral $\int_a^b f(x) - g(x) dx$ is always non-negative.

(d) The integral $\int_a^b |f(x) - g(x)| dx$ represents the area between the two curves.

Revision (1/1). The answers (a) and (d) are accurate, while (b) and (c) are not. For (a), areas are always non-negative so that is easy. For (d), this integral is how you calculate the area (we learned this in the reading on pg. 109). Now we deal with (b) and (c). Suppose $f(x) = x$ and $g(x) = 2$ over the interval $[0, 1]$. Then

$$\int_0^1 x - 2 dx = \frac{1}{2} - 2 = -1.5.$$

So, (c) fails. The main issue is $g(x) = 2$ lies above $f(x) = x$ on $[0, 1]$ and so $\int_0^1 f(x) - g(x) dx$ calculates the *negative* of the area in this case. This also explains why (b) is false: the integral should have an absolute value symbol in it.

Tip: To show something is *false*, you can also provide a counter-example like we do here for cases (b) and (c).

Read. Check Revisions

Now let me illustrate how you may not get full points back on Reading Check revisions.

Question. (Multiple choice) Which of the following indefinite integrals are calculated accurately?

(a) $\int \sin(x) dx = \cos(x) + C.$

(b) $\int \cos(x) dx = \sin(x) + C.$

(c) $\int \tan(x) dx = \sec^2(x) + C.$

Revision (0/1). The integral (c) is accurate because I looked it up on pg. 431.

Logic: The revision is just wrong. You will know the answers before you revise, so I do not give credit for wrong answers, no matter the justification.

Question. (Select all that apply) Which of the following indefinite integrals are calculated accurately?

(a) $\int \cos(x) dx = \sin(x) + C.$

(b) $\int \sin(x) dx = \cos(x) + C.$

(c) $\int \tan(x) dx = \sec^2(x) + C.$

Revision (0.5/1). The integral (a) is accurate. The answer (a) is true because I looked it up on pg. 431 of our book.

Logic. The revision is partially right, but this is a select all question and so the revision process asks the student to also explain how they know (b) and (c) are not right answers.

Question. (Select all that apply) Suppose $y = f(x)$ and $y = g(x)$ are two curves lying over x in $[a, b]$. Which of the following statements are accurate?

(a) The area between the two curves is always non-negative.

(b) The integral $\int_a^b f(x) - g(x) dx$ represents the area between the two curves.

(c) The integral $\int_a^b f(x) - g(x) dx$ is always non-negative.

(d) The integral $\int_a^b |f(x) - g(x)| dx$ represents the area between the two curves.

Revision (0.5/1). The answers (a) and (d) are accurate, while (b) and (c) are not. For (a), areas are always non-negative so that is easy. For (d), this integral is how you calculate the area (we learned this in the reading on pg. 109). The reason (b) is wrong is because the absolute value in (d) needs to be there. The reason (c) is wrong is that the integral could be negative, if the area between the curves was negative.

Logic. The issue with this revision is that final phrase "...if the area between the curves was negative." That just is not an accurate statement, since area must be non-negative. I will read the revisions closely and parts of the explanation that are off like this will lead to less than full marks. (This is also why I recommend as on pg. 12 that you try to provide concrete examples to disprove statements.)

Perf. Task Revisions

Now let me illustrate revising a Performance Task might go. If you have any questions when revising your work, please come and find me or email me to ask your questions!

Question.

Determine the area between the curve $y = \sin(x)$ and $y = \cos(x)$ over the x -range of $[0, \pi/4]$.

Original Solution #1.

$$\cos(x) \geq \sin(x)$$

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = \sin(x) + \cos(x) \Big|_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414...$$

Revision.

Original score: 5/6.

New solution. When x is in $[0, \pi/4]$ we have $\cos(x) \geq \sin(x)$. To calculate the area, we therefore subtract the larger one from the smaller and determine the integral over $[0, \pi/4]$. What we get is

$$\int_0^{\pi/4} \cos(x) - \sin(x) \, dx = \sin(x) + \cos(x) \Big|_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414...$$

And therefore the area is 0.414.

Grader note: This revision would earn 6/6. And I am writing a few more words than you may, but the point is that the various steps have to be connected with some kind of words so that the grader can easily follow your work. Note also...please include your original score!

Original Solution #4. The area is given by

$$\sin(\pi/4) - \cos(\pi/4) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0.$$

Revision.

Original score: 3/6.

New solution. First, I should take the different in the right order. We have first

$$\cos(\pi/4) - \sin(\pi/4) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0.$$

I forgot to also calculate $\cos(0) - \sin(0) = 1$. Then I take the second answer and subtract the first. I get the area is 1.

Grader note: This revision would earn 4/6. The solver has included conceptual information that shows they understand there is something going on with calculating differences of functions. However, conceptual and mechanical gaps remain. In particular, there is no integral still, so the calculations are far too simple. Note: you may not earn back full points on Performance Tasks. Note also...please include your original score!

Generative AI Usage



Generative AI tools such as ChatGPT are powerful tools for learning. In my course, you are encouraged to responsibly use those tools to better your understanding.

Basic expectations

- **Reading Checks.** Complete Reading Checks *without* generative AI.
- **Exercise Sets.** Complete Exercise Sets *without* generative AI.
- **Mastery Assessments.** Complete revisions *without* generative AI.

Details. Learning is messy. It has neither start nor finish. It is just one tool you may use to better understand the material, and it's not like its going away. I've used it to help me study German, for instance. Students often use it to summarize or reexplain difficult ideas from their lessons. If you do that in this course, let me know what you find! I designed this course knowing about generative AI capabilities. The revision system, in particular, is meant to relieve the pressure some you may feel to take learning shortcuts.



Your "at home" work consists of Reading Checks, Exercise Sets, and Mastery Assessment revisions. In the next two columns I describe fair and unfair use of generative AI, in this course.

Fair use

- You **may** use generative AI before or after reading to reinforce your understanding.
- You **may** use generative AI to help explain a Reading Check question after it has been graded.
- You **may** use generative AI to generate examples similar to Exercise Set problems.
- You **may** use generative AI to help you find mistakes in the graded work (Exercise Sets or Mastery Assessments) that we return.
- You **may** model your language usage on the usage of a generative AI assistant.

Unfair use

- You **may not** directly ask generative AI to answer your Reading Check questions.
- You **may not** use generative AI to generate explanations you then turn in as revisions to Reading Check questions.
- You **may not** directly ask generative AI to answer Exercise Set problems for turn in.
- You **may not** use generative AI to generate explanations you then turn in as revisions to Exercise Sets or Mastery Assessments.
- You **may not** directly copy outputs from generative AI assistants.

Generative AI, cont.



The purpose of this page to provide a warning that using generative AI in my grading scheme may lead only to B-level grades, and that is only for the 35% of your grade based on Reading Checks and Exercise Sets. The learning gap AI creates with frequent use for shortcuts will not be worth it. Consider the question we used on p. 10-11.

Question.

Determine the area between $y = \sin(x)$ and $y = \cos(x)$ over x in $[0, \pi/4]$.

Answer from Chat GPT (prompt given Aug 3, 2025, edited for formatting).

We will do this in four steps.

1. Determine which function is on top.

- At $x = 0$, we have $\sin(0) = 0$ and $\cos(0) = 1$, so $\cos(x)$ is on top.
- At $x = \pi/4$ we have $\sin(\pi/4) = \cos(\pi/4) = \frac{\sqrt{2}}{2}$, so they are equal.

Therefore $\cos(x)$ is on top of $\sin(x)$.

2. Set up the area integral. We have $(\text{Area}) = \int_0^{\pi/4} \cos(x) - \sin(x) dx$.

3. Solve the integral. We have

$$\int_0^{\pi/4} \cos(x) - \sin(x) dx = \sin(x) + \cos(x) \Big|_0^{\pi/4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} - (0 + 1) = \sqrt{2} - 1.$$

4. The answer is $(\text{Area}) = \sqrt{2} - 1$.

Grading this solution: 5/6.

- *Concept.* 2/2. The solver orders the two functions, convert the area into an integral and then solve the integral.
- *Mechanics.* 1/2. There is a mechanical gap in Step #1. The solver only checks that $\cos(x) \geq \sin(x)$ for the two endpoints $x = 0$ and $x = \pi/4$. The mechanics says you need to justify it for all values of x , otherwise you risk that $y = \cos(x)$ crosses $y = \sin(x)$ somewhere in the middle of the range.
- *Grade-ability:* 1/1. The writing (typing) is neat.
- *Presentation:* 1/1. The writing narrates in a clear way how the question is being answered and what the outcome is.

Notes. The example question was on my syllabi last year also, but the AI pages were added this year. I did not "rig" the example to create a gap. ChatGPT did that all on its own! I have seen gaps like this a lot when asking ChatGPT questions starting at the 2xxxx-level in the University curriculum. The primary flaw in overusing generative AI is mistaking its confidence for correctness. Generative AI tools lack true intelligence and logical or learning capabilities. They are language models that guess answers based on statistics and probability. They do not use deductive reasoning and they do not reference or check background knowledge. They are often wrong about basic work and will give you absolutely no indication there is a gap.